08 — Feedback System & Learning Signals

Overview

The feedback system closes the memory loop: symbolic feedback labels from the user are converted into numeric learning signals that drive salience updates, confidence adjustments, and review flags on schemas.

Feedback Signal

Each user feedback label maps to a `FeedbackSignal` vector with 10 components:

$$\phi = (v, c_f, e_t, e_\tau, m, o, \Delta_s, \Delta_c, r_p, r_o)$$

Component	Symbol	Range	Description
`valence`	v	[−1, +1]	Overall usefulness (dopaminergic reward)
`context_fit`	c_f	[−1, +1]	Match between memory and query cue
`truth_error`	e_t	[0, 1]	Factual wrongness (prediction error)
`temporal_error`	e_τ	[0, 1]	Staleness / outdatedness
`missingness`	m	[0, 1]	Recall gap (needed info not retrieved)
`overload`	o	[0, 1]	Working memory capacity failure
`salience_delta`	Δ_s	$\mathbb{R}$	Absolute change to schema salience
`confidence_delta`	Δ_c	$\mathbb{R}$	Absolute change to schema confidence
`review_pressure`	r_p	[0, 1]	Urgency for manual review
`outcome_reward`	r_o	[−1,	Task-level reward (separate from memory

Component	Symbol	Range	Description
		+1]	quality)

Feedback Label Mapping

`useful`

$$\phi_{\text{useful}} = (1.0, 1.0, 0, 0, 0, 0, \Delta_s^{\text{use}}, \Delta_c^{\text{use}}, 0, r_o)$$

- Schema salience: `+0.15`, confidence: `+0.05`

`partially_useful`

$$\phi_{\text{partial}} = (0.4, 0.5, 0, 0, 0, 0, \Delta_s^{\text{part}}, 0, 0, r_o)$$

- Schema salience: `+0.05`, confidence: 0

`irrelevant`

$$\phi_{\text{irrel}} = (-0.4, -1.0, 0, 0, 0, 0, \Delta_s^{\text{irr}}, 0, 0, r_o)$$

- Schema salience: `-0.10`

`stale`

$$\phi_{\text{stale}} = (-0.6, -0.3, 0, 1.0, 0, 0, \Delta_s^{\text{stale}}, \Delta_c^{\text{stale}}, r_p^{\text{stale}}, r_o)$$

- Schema salience: `-0.20`, confidence: `-0.15`, review pressure: `0.6`

`wrong`

$$\phi_{\text{wrong}} = (-1.0, -0.5, 1.0, 0, 0, 0, \Delta_s^{\text{wrong}}, \Delta_c^{\text{wrong}}, r_p^{\text{wrong}}, r_o)$$

- Schema salience: `-0.30`, confidence: `-0.30`, review pressure: `0.8`

`missing`

$$\phi_{\text{missing}} = (-0.3, 0, 0, 0, 1.0, 0, 0, 0, 0, r_o)$$

- No salience/confidence delta — missingness flags that something should have been retrieved but wasn't.

`too_much_context`

$$\phi_{\text{overload}} = (-0.2, -0.2, 0, 0, 0, 1.0, 0, 0, 0, r_o)$$

- No salience/confidence delta — overload is a gating problem, not a memory quality problem.

Outcome Reward

The outcome label (`success`, `partial`, `unknown`, `failure`) maps independently:

Outcome	r_o
`success`	`+1.0`
`partial`	`+0.3`
`unknown`	`0.0`
`failure`	`-1.0`

By default, outcome reward is **not** applied to schema reward (`apply_outcome_to_schema_reward = False`) — outcome measures task success, not memory quality.

Schema Updates from Feedback

For each schema identified in `used_memory_ids`:

$$s_{\text{schema}} \leftarrow s_{\text{schema}} + \Delta_s$$

$$c_{\text{schema}} \leftarrow c_{\text{schema}} + \Delta_c$$

For stale/wrong schemas, `needs_review` is set when `review_pressure ≥ stale_review_threshold` or `wrong_review_threshold`.

For `irrelevant`, `stale`, `wrong` schemas, the `context_noise_score` is incremented — this feeds into the working-memory gate's noise penalty.

Configuration

`FeedbackConfig`

Parameter	Type	Default	Description
`enabled`	`bool`	`True`	Master enable
`persist_context_snapshots`	`bool`	`True`	Store context recall events
`persist_response_json`	`bool`	`True`	Store response metadata
`persist_rendered_context`	`bool`	`False`	Store rendered text (can be large)
`persist_activation_trace`	`bool`	`False`	Store activation trace (very large)
`max_response_json_chars`	`int`	`20000`	Truncation for response JSON
`max_memory_content_chars`	`int`	`500`	Truncation for memory content
`apply_learning`	`bool`	`True`	Master learning gate
`apply_positive_learning`	`bool`	`True`	Enable positive reinforcement
`apply_negative_learning`	`bool`	`True`	Enable negative feedback
`apply_stale_wrong_review`	`bool`	`True`	Enable review flagging
`apply_outcome_to_schema_reward`	`bool`	`False`	Map outcome to schema reward

Parameter	Type	Default	Description
`context_feedback_weight`	`float`	`0.5`	Weight for context-level feedback
`useful_salience_delta`	`float`	`0.15`	Salience boost for `useful`
`useful_confidence_delta`	`float`	`0.05`	Confidence boost for `useful`
`partially_useful_salience_delta`	`float`	`0.05`	Salience boost for `partially_useful`
`irrelevant_salience_delta`	`float`	`-0.10`	Salience penalty for `irrelevant`
`stale_salience_delta`	`float`	`-0.20`	Salience penalty for `stale`
`stale_confidence_delta`	`float`	`-0.15`	Confidence penalty for `stale`
`stale_review_threshold`	`float`	`0.6`	Review pressure threshold for `stale`
`wrong_salience_delta`	`float`	`-0.30`	Salience penalty for `wrong`
`wrong_confidence_delta`	`float`	`-0.30`	Confidence penalty for `wrong`
`wrong_review_threshold`	`float`	`0.8`	Review pressure threshold for `wrong`

Key Invariants

1. `missing` and `too_much_context` produce no schema updates — they are retrieval/gating issues, not memory quality issues.
2. Outcome is tracked separately from memory quality — a task can succeed despite irrelevant context.
3. Salience deltas are small additive changes (`±0.05` to `±0.30`) — feedback accumulates over many sessions.

4. Review flags are irreversible until manual intervention.